Woods & manufactured boards

D&M WITH MR KELLY

Higher · Materials & manufacture · Worksheet

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ describe the properties and uses of beech, ash, oak and mahogany
- ☐ describe the properties and uses of red pine and spruce
- ☐ describe the properties and uses of plywood, flexi-ply, MDF, chipboard and hardboard
- ☐ describe blockboard and veneered boards

I DO — Worked example

Read how a full-mark answer is built

A flat-pack bedside cabinet has a carcass of veneered MDF and a solid oak top.

1 Explain why veneered MDF is suitable for the carcass. (3 marks)

Points that earn marks from the marking instructions: one ✓ per point

- ✓ Veneer gives the look of a quality hardwood at lower cost
- ✓ MDF is stable, so the panels will not warp
- ✓ Comes in large flat sheets, so each panel is one piece

Why marks are not awarded

- ✗ "Cheap" or "strong" with no explanation

WE DO — Guided practice

Use the starters and word bank to help you

A flat-pack bedside cabinet has a carcass of veneered MDF and a solid oak top.

2 Describe two advantages and one disadvantage of using solid oak for the top. (3 marks)

Sentence starters

One … is …, which means …
Another … is …, so …
Also, … because …

Word bank

resists · price · advantage · scratches · attractive · durable · warp · knocks

Tip: 3 marks = 3 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) Which board would you use for a curved chair back?
☐ Chipboard ☐ Hardboard ☐ Flexi-ply ☐ MDF

b) Why does plywood have an odd number of layers with grain at 90°?
☐ To make it cheaper ☐ To reduce warping and add strength ☐ So it bends easily ☐ For colour

c) Which hardwood suits a cricket bat or hammer handle?
☐ Ash ☐ Mahogany ☐ MDF ☐ Spruce

d) Which is a softwood?
☐ Beech ☐ Oak ☐ Spruce ☐ Mahogany

4 Key terms: write the letter of the matching description next to each term.

1. **Softwood source**	____	**A** Hardwood: very strong, turns well. Toys, spoons, chairs.
2. **Hardwood source**	____	**B** Odd number of veneers with grain at 90°: strong and stable.
3. **Beech**	____	**C** Hardwood: tough, bends well. Sports bats, tool handles.
4. **Ash**	____	**D** Hardwood: very durable. Quality and outdoor furniture, barrels.
5. **Oak**	____	**E** Coniferous trees: evergreen, needle leaves, fast growing.
6. **Plywood**	____	**F** Deciduous trees: broad leaves shed in winter, slow growing.

5 Give the properties and one use of MDF. (2 marks)

6 Describe the advantage of manufactured boards. (2 marks)

Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?